

Generative AI & Agentic AI for Network and Infrastructure Engineering

Duration: 5 Days | 4 Hours per Day | 20 Hours

Level: Intermediate to Advanced

Delivery Mode: Instructor-Led | Demonstrations | Hands-on Labs | Architecture Exercises | Enterprise Network & Infrastructure Scenarios

Course Objectives

By the end of this program, participants will be able to:

- Understand Generative AI, LLMs, reasoning models and Agentic AI in the context of Network and Infrastructure Engineering.
- Apply GenAI to network logs, alerts, configurations, telemetry, incidents and operational troubleshooting.
- Engineer effective prompts and structured outputs for NetOps and InfraOps use cases.
- Perform evidence-driven incident investigation and AI-assisted root-cause analysis.
- Use GenAI to accelerate Python, Ansible, Terraform and infrastructure automation.
- Build enterprise RAG solutions grounded in runbooks, SOPs, standards and historical incidents.
- Integrate AI applications with operational systems using tool/function calling and Model Context Protocol (MCP).
- Understand model adaptation and determine when to use prompting, RAG, tool calling or fine-tuning.
- Design tool-enabled AI agents for multi-step network and infrastructure operations.
- Design single-agent and multi-agent architectures for Agentic NetOps.
- Implement human-in-the-loop and bounded-autonomy patterns for controlled remediation.
- Design closed-loop workflows covering detection, investigation, remediation and validation.
- Evaluate AI systems for RCA accuracy, groundedness, tool-use accuracy, safety, latency and cost.
- Apply enterprise security, governance, observability and responsible-AI controls to production GenAI systems.
- Architect scalable, reliable and secure GenAI/Agentic AI solutions for enterprise Network and Infrastructure Operations.

Course Prerequisites

- Working knowledge of Network, Infrastructure, Cloud, Platform, SRE or IT Operations concepts.
- Familiarity with logs, alerts, configurations, incidents and troubleshooting workflows.
- Basic understanding of APIs and infrastructure automation.
- Basic programming/scripting knowledge, preferably Python.
- Exposure to JSON/YAML, Ansible, Terraform or REST APIs is helpful but not mandatory.

- No prior Generative AI, LLM or Agentic AI experience is required.

Course Content

Day 1 — Generative AI & Reasoning Foundations for Network and Infrastructure Engineers

Module 1: Generative AI in Modern Network & Infrastructure Operations

- Evolution from traditional automation and AIOps to Generative AI
- AI vs ML vs GenAI vs Agentic AI
- Enterprise NetOps and InfraOps use cases
- Copilots, assistants and AI agents
- Deterministic automation vs probabilistic AI systems
- Where GenAI adds value and where traditional automation remains appropriate

Module 2: LLM & Reasoning Model Essentials

- Foundation models and Large Language Models
- Tokens, context windows and inference
- Transformer concepts for infrastructure engineers
- Standard LLMs vs reasoning models
- Model capabilities and limitations
- Hallucinations, uncertainty and evidence-based reasoning
- Latency, quality and cost trade-offs

Module 3: Prompt & Context Engineering for Infrastructure Operations

- System prompts and user prompts
- Zero-shot and few-shot prompting
- Role, task, context and constraints
- Context engineering
- Structured outputs and JSON responses
- Prompt patterns for troubleshooting
- Evidence-based prompting
- Controlling hallucinations and unsupported conclusions

Module 4: Applying GenAI to Infrastructure Data

- Network and infrastructure logs
- Syslogs and CLI outputs
- Device configurations
- Monitoring alerts
- Telemetry summaries
- JSON/YAML
- Incident tickets
- Runbooks and SOPs

- Configuration changes

Hands-on Lab 1: AI Network Log Detective

Analyze operational logs and transform them into:

Symptoms → Evidence → Possible Causes → Investigation Steps → Recommended Action

Hands-on Lab 2: Build an Infrastructure Troubleshooting Copilot

Build a GenAI-powered assistant that analyzes incident context and produces a structured investigation report.

Day 1 Outcome

Understand → Prompt → Analyze → Reason

Day 2 — GenAI-Assisted NetOps, Root-Cause Analysis & Infrastructure Automation

Module 1: AI-Assisted Network & Infrastructure Troubleshooting

- Connectivity failures
- Routing and interface issues
- DNS/DHCP problems
- Latency and packet-loss scenarios
- Firewall/connectivity problems
- Configuration inconsistencies
- Infrastructure availability issues
- AI-assisted troubleshooting workflows

Module 2: Intelligent Log, Alert & Event Analysis

- Log summarization
- Alert enrichment
- Event correlation
- Alert-noise reduction
- Incident clustering
- Timeline reconstruction
- Converting operational events into engineering narratives

Module 3: Evidence-Driven Root-Cause Analysis

- Multi-source evidence collection
- Hypothesis generation

- Hypothesis ranking
- Correlation vs causation
- Evidence supporting and contradicting hypotheses
- Confidence and uncertainty
- Preventing hallucinated RCA
- Human validation of AI conclusions

Hands-on Lab 3: AI Incident War Room

Investigate an enterprise network incident using:

Alerts + Logs + Telemetry + Configuration + Recent Changes + Incident Context

Identify and defend the most probable root cause using evidence.

Module 4: GenAI for Infrastructure Automation

- AI-assisted Python development
- Shell/PowerShell assistance
- Ansible generation and explanation
- Terraform assistance
- YAML/JSON generation
- REST API integration
- Configuration templates
- Infrastructure-as-Code assistance
- Safe generation and validation patterns

Hands-on Lab 4: Prompt-to-Automation Challenge

Convert a natural-language infrastructure requirement into an automation workflow and apply:

Generate → Explain → Validate → Test → Approve

Module 5: GenAI for Operational Productivity

- Incident summaries
- Change-plan generation
- Troubleshooting documentation
- Runbook creation
- Post-incident/RCA reports
- Configuration review assistance
- Knowledge capture from resolved incidents

Day 2 Outcome

Investigate → Correlate → Diagnose → Automate

Day 3 — Enterprise RAG, Tool Calling, MCP & Model Adaptation

Module 1: Grounding AI in Enterprise Infrastructure Knowledge

- Why generic LLM knowledge is insufficient
- Enterprise-specific network context
- Internal standards and policies
- Runbooks and troubleshooting knowledge
- Configuration standards
- Historical incidents
- Operational knowledge boundaries

Module 2: Retrieval-Augmented Generation (RAG) for NetOps

- RAG architecture
- Knowledge ingestion
- Chunking strategies
- Embeddings
- Vector databases
- Semantic search
- Retrieval strategies
- Context augmentation
- Grounded generation
- Source attribution
- RAG quality and failure modes

Hands-on Lab 5: Ask My Network Runbooks

Build an enterprise RAG assistant grounded in:

Runbooks + SOPs + Network Standards + Troubleshooting Guides + Historical Incidents

Module 3: RAG Beyond Documents

- Configuration repositories
- CMDB
- Network topology
- Incident history
- Change records
- Monitoring information
- Structured vs unstructured retrieval
- Hybrid enterprise knowledge retrieval

Module 4: Tool/Function Calling for Infrastructure AI

- From conversational AI to actionable AI

- Function/tool definitions
- Tool selection
- Parameter generation
- Tool-response processing
- Tool failures and retries
- Permission boundaries
- Safe infrastructure tool execution

Example capabilities:

```
get_device_health()
get_interface_status()
query_monitoring()
retrieve_configuration()
get_recent_changes()
search_incidents()
run_diagnostic()
```

Hands-on Lab 6: Give the AI Network Tools

Build an AI assistant capable of selecting and invoking approved diagnostic tools to investigate an infrastructure problem.

Module 5: Model Context Protocol (MCP) for Infrastructure Engineering

- Why MCP matters for enterprise AI
- MCP architecture
- MCP clients and servers
- Tools, resources and prompts
- MCP vs direct API integration
- Exposing infrastructure capabilities through MCP
- Authentication and authorization
- MCP security and permission boundaries

Hands-on Lab 7: MCP-Powered Infrastructure Diagnostics

Expose or consume safe diagnostic capabilities through MCP and integrate them with an AI workflow.

Module 6: Model Adaptation & Fine-Tuning

- Prompt engineering vs RAG vs tool calling vs fine-tuning
- When fine-tuning is justified
- Domain adaptation
- LoRA/QLoRA concepts
- Synthetic network incident datasets
- Fine-tuning for specialized behavior and structured outputs
- Evaluation before and after adaptation
- Cost and operational considerations

Day 3 Outcome

Retrieve → Ground → Integrate → Adapt

Day 4 — Agentic AI & Multi-Agent Network Operations

Module 1: From GenAI Copilot to Agentic AI

- LLM vs chatbot vs copilot vs AI agent
- Agent goals and instructions
- Reasoning and planning
- Tools
- Memory and state
- Observations and actions
- Feedback loops
- Agentic workflow patterns

Module 2: Agentic Network Investigation

Implement the operational reasoning loop:

Observe → Reason → Plan → Act → Validate

Typical workflow:

Incident → Collect Evidence → Query Monitoring → Inspect Configuration → Check Changes → Retrieve Knowledge → Generate Hypotheses → Validate → Recommend

Hands-on Lab 8: Build an AI Network Troubleshooting Agent

Build a tool-enabled agent that autonomously determines:

- What evidence is required
- Which diagnostic tool to invoke
- What additional context is missing
- Which hypotheses are supported
- What action should be recommended

Module 3: Agent Memory, State & Context Management

- Short-term memory
- Persistent memory
- Workflow state
- Incident context
- Memory boundaries
- Stale context risks
- State management across multi-step investigations

Module 4: Multi-Agent Systems for Network & Infrastructure Operations

Design specialized agents such as:

- Incident Coordinator Agent
- Monitoring/Telemetry Agent
- Topology Agent
- Configuration Agent
- Knowledge/RAG Agent
- RCA Agent
- Security/Policy Agent
- Remediation Agent
- Validation Agent
- Incident Documentation Agent

Module 5: Single Agent vs Multi-Agent Architecture

- When one agent with multiple tools is sufficient
- When specialized agents add value
- Agent delegation and handoffs
- Agent-to-agent communication
- Coordination challenges
- Reliability and failure propagation
- Latency and cost considerations

Hands-on Lab 9: Design an Agentic NOC

Teams design a multi-agent incident-management workflow and justify:

Agent Boundaries → Tools → Handoffs → Permissions → Failure Handling → Human Controls

Module 6: Human-in-the-Loop & Controlled Remediation

- AI recommendation vs AI execution
- Approval workflows
- Least-privilege execution
- Bounded autonomy
- Policy-driven actions
- Validation and rollback
- Escalation to engineers

Hands-on Lab 10: Should the Agent Execute?

Evaluate remediation scenarios and determine:

Recommend → Approve → Execute → Validate → Rollback/Escalate

Day 4 Outcome

Reason → Plan → Collaborate → Act → Validate

Day 5 — Autonomous Operations, Production Architecture, Security, Governance & Evaluation

Module 1: From Agentic AI to Closed-Loop Operations

- Reactive vs proactive operations
- AI-assisted vs autonomous operations
- Closed-loop remediation
- Self-healing infrastructure concepts
- Event-driven agent activation
- Continuous verification

Operational loop:

Detect → Investigate → Diagnose → Plan → Approve → Execute → Validate → Rollback/Escalate

Module 2: Designing Safe Levels of AI Autonomy

- Manual operations
- AI-assisted recommendations
- Human-approved execution
- Bounded autonomous execution
- Policy-controlled closed-loop operations
- Defining autonomous-action boundaries
- Blast-radius considerations

Module 3: Production GenAI & Agentic AI Architecture

- Model gateway
- RAG/knowledge layer
- Tool/MCP layer
- Agent orchestration
- API and integration layer
- Event-driven architecture
- Model selection and routing
- Hosted vs private models
- Scalability and resilience
- Caching and context management
- Model fallback
- Cost and latency optimization

Module 4: Security for AI-Enabled Infrastructure Operations

- Prompt injection
- Data leakage

- Sensitive configuration exposure
- Secrets and credential protection
- Unsafe generated commands
- Excessive agent permissions
- Unauthorized infrastructure changes
- Tool abuse
- RAG/knowledge poisoning
- MCP security considerations
- Least privilege for AI agents

Hands-on Lab 11: Red-Team the Infrastructure Agent

Identify and mitigate:

Threat → Attack Path → Infrastructure Impact → Guardrail → Safe Agent Response

Module 5: Enterprise Guardrails & Governance

- Role-based access
- Tool allowlists
- Command restrictions
- Policy engines
- Approval workflows
- Output validation
- Change-management integration
- Auditability
- Rollback mechanisms
- Separation of recommendation and execution

Module 6: Observability for GenAI & Agentic Systems

Trace the complete workflow:

Prompt → Retrieval → Reasoning → Tool Calls → Agent Actions → Approval → Execution → Result

Monitor:

- Traces and spans
- Agent decisions
- Tool calls
- Failures and retries
- Latency
- Token consumption
- Cost
- Approvals
- Infrastructure actions

Module 7: Evaluating Network & Infrastructure AI

- Golden incident datasets
- RCA accuracy
- Evidence groundedness
- Hallucination rate
- Retrieval quality
- Tool-selection accuracy
- Tool-argument accuracy
- Task-completion rate
- Remediation accuracy
- False-action rate
- Safety violations
- Latency and cost
- Regression testing

Module 8: Synthetic Data for Network AI

- Generating synthetic incidents
- Synthetic syslogs and alerts
- Configuration-error scenarios
- Telemetry anomalies
- Rare failure scenarios
- Synthetic datasets for RAG, fine-tuning and agent evaluation

Module 9: Network Infrastructure for GenAI Workloads

- Infrastructure requirements of GenAI systems
- Training vs inference workloads
- GPU infrastructure fundamentals
- East-west traffic characteristics
- High-bandwidth/low-latency networking
- AI cluster considerations
- Storage and network considerations
- Scaling inference workloads
- Cloud vs on-premises AI infrastructure
- Observability and availability considerations

Final Capstone — AI-Powered Agentic Network Operations & Incident Investigation System

Participants design/build an end-to-end Agentic AI solution for a business-critical network/infrastructure incident.

Capstone Workflow

Incident Detection



Alert & Telemetry Analysis



AI Agent Investigation



Network/Infrastructure Tool Calls



Configuration & Change Analysis



Enterprise RAG / Runbook Retrieval



Historical Incident Correlation



Root-Cause Hypothesis Generation



Evidence-Based Validation



Remediation Planning



Security & Policy Validation



Human Approval / Bounded Autonomy



Controlled Remediation



Service Validation



Rollback or Escalation



Automated RCA & Incident Closure Report

Capstone Architecture Challenge

Participants define and defend:

- What should AI **observe**?
- What should AI **reason about**?
- What tools should AI be permitted to **access**?
- What should AI only **recommend**?
- What actions should require **human approval**?
- What low-risk actions can be **autonomous**?
- What should AI **never be permitted to execute**?
- How will actions be **validated, audited and rolled back**?